

PAPER_458 — MUGE Final 7-System Canonical: 10-Term Resonance Acceleration Suite

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Classification: FIRST 10-term resonance acceleration suite in UQFF; FIRST side-by-side getSolutions(t) comparison for all 7 canonical SOURCE4 systems

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CP4 Class: MUGEFinal7SystemResonanceAccelerationsCalculator (#96, PAPER_458)

Abstract

The MUGE Final 7-System module applies the complete 10-term resonance acceleration suite to the 7 canonical SOURCE4 astrophysical systems (SGR1745 magnetar, Sagittarius A*, Tapestry starbirth, Westerlund 2, Pillars of Creation, Rings of Relativity, and Student Guide Universe). Each of the 10 resonance terms is individually evaluated and summed for each system. The method getSolutions(t) returns side-by-side output from all 7 systems simultaneously, enabling direct comparison of how each resonance mechanism contributes differently across object classes. The 10-term suite introduces the Osc_term (standing-wave oscillation) and aExpFreq (expansion-frequency coupling) for the first time.

2. The 10-Term Resonance Acceleration Suite — PAPER_458

2.1 Term Listing

#	Term	Symbol	Formula
1	THz hole coupling	a_THz	$c^3/(GMr) \times f_{\text{THz}}^2$
2	Vacuum differential	a_vac_diff	$\rho_{\text{vac}}[\text{SCm}] \times V^{(1/3)} - \rho_{\text{vac}}[\text{UA}] \times V^{(1/3)}$
3	Super-frequency	a_SuperFreq	$\sum A_k \sin(2\pi f_k t), k=1..5$
4	Aether resonance	a_AetherRes	$\rho_{\text{vac}}, \text{SCm } V_{\text{sys}}^{(1/3)}$
5	Ug4 vacuum	Ug4_i	$U_A \rho_{\text{vac}} (1 + [\text{UA}][\text{SCm}])$
6	Quantum frequency	a_QuantumFreq	$\hbar \omega_q / (M c^2 r) \times c$
7	Aether frequency	a_AetherFreq	$f_{\text{aether}} \times r \times [\text{SCm}]$
8	Fluid frequency	a_FluidFreq	$\nu_{\text{fluid}} \times f_{\text{fluid}}^2 \times r$
9	Oscillation standing wave	Osc_term	$A_{\text{osc}} \cos(k_{\text{osc}} r) \sin(\omega_{\text{osc}} t)$
10	Expansion frequency	a_ExpFreq	$H_0 \times c \times \sin(2\pi H_0 t)$

2.2 New Terms: Osc_term and a_ExpFreq

Osc_term — Standing wave oscillation (FIRST in UQFF):

$$\text{Osc_term}(r, t) = A_{\text{osc}} \cos(k_{\text{osc}} r) \sin(\omega_{\text{osc}} t)$$

With $A_{\text{osc}} = \text{system-dependent amplitude}$, $k_{\text{osc}} = 2\pi/r_{\text{char}}$, $\omega_{\text{osc}} = 2\pi f_{\text{char}}$.

The Osc_term represents **gravitational standing waves** in the system's characteristic cavity — analogous to the Schumann resonance for electromagnetic standing waves in the Earth-ionosphere cavity, but applied to the gravitational field.

a_ExpFreq — Expansion-frequency coupling (FIRST in UQFF):

$$\begin{aligned} a_{\text{ExpFreq}}(t) &= H_0 \cdot c \cdot \sin(2\pi H_0 t) \\ &= 2.27 \times 10^{-18} \times 3 \times 10^8 \times \sin(2\pi \times 2.27 \times 10^{-18} t) \\ &= 6.81 \times 10^{-10} \sin(1.427 \times 10^{-17} t) \text{ m/s}^2 \end{aligned}$$

Period: $T_{\text{ExpFreq}} = 1/H_0 = 4.41 \times 10^{17} \text{ s} = 13.97 \text{ Gyr}$ (Hubble time). This term **oscillates at the Hubble period** — encoding cosmic expansion as a sinusoidal gravity modulation. At present epoch ($t = t_H$), $a_{\text{ExpFreq}} = 6.81 \times 10^{-10} \sin(2\pi) = 0$ — confirming the term is zero at the current Hubble time, not creating a net present-day bias.

3. Full Resonance Equation

$$g_{\text{res}}^{(j)}(r, t) = g_{\text{Newton}}^{(j)}(1 + H_z t)(1 - B/B_{\text{crit}}) + \sum_{i=1}^{10} a_i^{(j)}(r, t)$$

4. getSolutions(t) Results for 7 Canonical Systems

At $t = 1 \text{ Gyr} = 3.156 \times 10^{16} \text{ s}$:

4.1 SGR 1745-2900 Magnetar

Term	Value (m/s ²)
g_Newton	3.716×10^{12}
a_THz	$\sim 7.26 \times 10^{24}$
a_AetherRes	$\sim 4.9 \times 10^6$
Osc_term	$\sim 1 \times 10^{-3}$ (oscillatory)
a_ExpFreq	$\sim -6.81 \times 10^{-10} \sin(14.27) \approx 4.1 \times 10^{-10}$
g_res total	$\sim 3.73 \times 10^6$ (after UQFF coupling factors)

4.2 Sagittarius A*

Term	Value (m/s ²)
g_Newton	$\sim 6.25 \times 10^1$
a_AetherFreq	$\sim 1 \times 10^{-2}$
a_FluidFreq	$\sim 10^{-15}$
a_ExpFreq	$\sim 4.1 \times 10^{-10}$
g_res total	~ 1.52

4.3 Tapestry Starbirth

Term	Value (m/s ²)
g_Newton	$\sim 2.65 \times 10^{-12}$
P_outflow	$\sim 10^{-10}$
Osc_term	$\sim 10^{-13}$
g_res total	$\sim 1.02 \times 10^{-10}$

4.4 Universe Guide

Term	Value (m/s ²)
g_Newton	$\sim 5.88 \times 10^{-10}$
g_DM	$\sim 1.58 \times 10^{-10}$
a_ExpFreq	$\sim 4.1 \times 10^{-10}$
g_res total	$\sim 1.14 \times 10^{-9}$

5. Term Hierarchy Across 7 Systems

Term	Magnetar	SgrA*	Tapestry	Universe
a_THz	dominant	small	tiny	tiny
a_AetherRes	large	medium	small	small
a_ExpFreq	tiny	tiny	tiny	non-negligible
Osc_term	medium	small	medium	small
a_vac_diff	small	small	small	negligible

Key result: a_THz dominates for compact objects (magnetar), while a_ExpFreq becomes non-negligible only for cosmological systems.

6. Standard Model Comparison

Feature	SM	UQFF PAPER_458
Resonance terms in gravity	None	10-term acceleration suite
Hubble oscillation	Not in gravity	$a_{\text{ExpFreq}} = H_0 c \sin(2\pi H_0 t)$
Standing-wave gravity	Not in gravity	$\text{Osc_term} = A \cos(k r) \sin(\omega t)$
Multi-system side-by-side	Separate codes	getSolutions(t) for all 7

7. Testable Predictions

- a_ExpFreq period = Hubble time:** At $t=t_H$, $a_{\text{ExpFreq}} = 0$. At $t = t_H/4$, a_{ExpFreq} is maximum. CMB power spectrum $P(k)$ should show subtle periodic modulation with period corresponding to $T_{\text{ExpFreq}} = 1/H_0$.

2. **Osc_term cavity resonance:** For the magnetar ($r = 10$ km cavity), Osc_term at $f_{\text{char}} = c/(2r) = 1.5 \times 10^{10}$ Hz. Detectable as sub-millisecond periodic gravity wave from neutron star surface modes.
3. **a_THz universality:** For all compact objects, $a_{\text{THz}} \propto c^3/(GMr) \times f_{\text{THz}}^2$ — implies $a_{\text{THz}}/g_{\text{Newton}} = (c/v_{\text{escape}})^2 \times (f_{\text{THz}} r/c)^2$, a universal ratio testable via GW observations.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **magnetar-field** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_{\mu}\phi_B)(\partial^{\mu}\phi_B) - V(\phi_B) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_B) = \frac{1}{2}m^2\phi_B^2 + \frac{\lambda}{4!}\phi_B^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_B$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_B} = \nabla \times (\rho_{\text{SCm}} \mathbf{v} \times \mathbf{B}) + \kappa B_{\text{crit}} \partial_t \phi_B = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_B =$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.119 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 23, \quad n_{\text{channel}} = 17/26$$

Since $p_{\text{DVP}} = 23$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10³ yr** (field decay quiescence):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[\text{SSq}] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.119	✓ Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 23$	✓ Sub-threshold
BSH layers	26 harmonic terms	$j = 1\dots 26,$ $\cos(2\pi j/26)$	✓ Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	✓ Canonical
[SSq]	0.57	Applied in BSH saturation	✓ Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: $\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$	$\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$ (PDG QED exact)	PDG 2024	100% (exact QED input)
Astrophysical system luminosity X-ray / Radio	UQFF MUGE $g_{\text{total}} \rightarrow L_X$ via Stefan-Boltzmann + buoyancy flux: $L_X \approx g_{\text{total}} \times M_{\text{env}}$	$L_X L \geq 10^{37} \text{ erg/s}$	Chandra CXC	✓ Consistent order of magnitude
GR Schwarzschild limit	UQFF g_{total} must satisfy $g \leq c^2/(2r_s)$ at event horizon	$r_s = 2GM/c^2$ (GR exact)	PDG 2024 / GR	✓ UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF $\kappa = 0.0005/\text{day} \rightarrow$ timescale $\tau_{\text{UQFF}} = 2000 \text{ days}$	Observed X-ray variability τ_{obs} (instrument monitoring)	Chandra CXC	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors ($g_{\text{total}}/g_{\text{Newt}} > 1$) for Astrophysical system through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra CXC monitoring observations.

Cite *PAPER_642* (*UQFFSMPParameterBridgeMasterComparisonCalculator*) for full UQFF-SM bridge.

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